

Generating serum-based antibodies from tumor-exposed mice: a new potential strategy for cutaneous squamous cell carcinoma treatment

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Abstract

Current cancer treatment strategies continue to face significant challenges, primarily due to tumor relapse, drug resistance, and low treatment efficiency. These issues arise because certain tumor cells adapt to the host immune microenvironment and evade the immune system. This study presents a new cancer immunotherapy strategy using serum-based antibodies from mice exposed to mouse cutaneous squamous cell carcinoma (mCSCC). The experiment was conducted in three stages. In the first stage, mCSCC cells were isolated and expanded cultured from DMBA/TPA-induced mCSCC. In the second stage, these expanded tumor cells were injected into healthy mice to stimulate the production of anti-tumor antibodies. In the final stage, therapeutic serum was extracted from these healthy mice and reintroduced into the tumor-bearing mice. An ELISA assay was utilized to analyze the levels of p53, Bcl-xL, NF- κ B, and Bax. The results showed that the serum treatment not only reduced tumor volume but also reversed changes in p53, Bcl-xL, NF- κ B, and Bax. In conclusion, this study developed a new immunotherapeutic strategy for treating mCSCC. However, further research is needed to fully comprehend the mechanism of this serum treatment.

eLife assessment

This study provides a **valuable** strategy for treating mouse cutaneous squamous cell carcinoma (mCSCC) with serum derived from mCSCC-exposed mice. The exploration of serum-derived antibodies as a potential therapy for curing cancer is particularly promising but the study provides **incomplete** evidence for specific effects of mCSCC-binding serum antibodies. This study will be of interest to scientists seeking a novel immunotherapeutic strategy in cancer therapy.

<https://doi.org/10.7554/eLife.95678.2.sa2>

1. Introduction

The primary challenges in cancer treatment today include cancer heterogeneity, therapeutic resistance, and tumor recurrence [1]. The predominant strategies for cancer therapy currently encompass chemotherapy, radiotherapy, and immunotherapy. Despite significant strides made in cancer treatment over the past decades, issues persist with resistance to traditional chemotherapeutic agents and a lack of specificity in targeting cells [2]. Certain cell surface proteins have emerged as valuable targets and biomarkers for cancer therapies. However, high tumor recurrence rates remain a significant concern. This is primarily due to the tumor cells express different biomarkers at different developmental stages [3]. Within a tumor mass, there coexists a multitude of tumor cells at different stages and of diverse types. Some of these cells can evade treatment targets when subjected to chemotherapy and radiotherapy that only target one or several biomarkers (mutated proteins) [4]. These evasive tumor cells, particularly the cancer stem cells with self-renewal and differentiation capabilities, undergo genetic alterations and modify cell-surface antigen production (mutated proteins) to evade the immune system [5]. Unexpectedly, these strategies may inadvertently create a conducive growth environment for these evaded tumor cells.

Immunotherapy is designed to strengthen the patient's immune system in order to eradicate tumor cells [6]. There are currently several types of immunotherapy utilized in cancer treatment, which include immune checkpoint inhibitors, T-cell transfer therapy, and monoclonal antibodies [7]. Despite significant improvements in both active and passive cancer immunotherapy over recent years, these methods have not completely succeeded in preventing the recurrence of tumors [8]. The primary reason is the ability of some tumor cells to adapt to the immune microenvironment and evade the immune system by altering the expression or structure of proteins, prevents immune cells from recognizing them as foreign antigens [9]. Furthermore, the aforementioned methods do not consider the tumor mass as a whole entity, which encompasses cancer cells at various developmental stages, each harboring a range of known and unknown mutated proteins. This implies that the pattern of tumor markers (mutated protein) associated with each individual's tumor is unique. As a result, these therapeutic approaches are unable to completely eradicate tumor cells across diverse types and stages. In this study, a new cancer treatment strategy is designed using mouse cutaneous squamous cell carcinoma (mCSCC) as a model. This strategy aims to treat tumors in three stages: isolating tumor cells, producing serum-based antibodies, and eliminating the tumor cells.

2. Materials and methods

2.1 DMBA/TPA carcinogenesis

Fifty C57BL/6 male mice were randomly divided into five groups: tumor + serum treatment, tumor + no serum treatment, control + serum treatment (control 1), control + no serum treatment (control 2), and serum provider. Each mouse from the tumor + serum treatment group was paired with a mouse of the same blood type (type A or type B) from the serum provider group. The mice in the tumor + serum treatment and tumor + no serum treatment groups received treatment with 7,12-Dimethylbenz(a)anthracene (DMBA) and 12-O-Tetradecanoylphorbol-13-acetate (TPA). The dorsal skin area of the mice was shaved. Two days later, the mice were topically treated with 60 µg of DMBA, dissolved in 200 µl of acetone, on their bare backs. This DMBA administration was carried out for two weeks, after which the mice were exposed to 2.5 µg of TPA in 200 µl of acetone once a week for a total of 10 weeks. DMBA (Lot: D3254) and TPA (Lot: P1585) were purchased from Sigma-Aldrich, China. Skin tumors were measured using a precision caliper, which allowed for the

detection of size changes greater than 0.1 mm. Body weights were recorded weekly. Tumor volumes were measured on the first day of treatment and every week thereafter until the end of the experiments. The volume was calculated using the formula $V = \pi \times [d^2 \times D]/6$, where V represents the volume of the tumor, d is the minor axis of the tumor (the shortest diameter), D is the major axis of the tumor (the longest diameter) [10]. **Figure 1** presents a workflow of this study. This study was performed in strict accordance with the recommendations in the Guide for the Care and Use of Laboratory Animals of the National Institutes of Health. All of the animals were handled according to approved institutional animal care and use committee protocols of Guilin Medical University. The protocol was approved by the Experimental Animal Ethics Committee of Guilin Medical University (Permit Number: GLMC202203177). All surgery was performed under sodium pentobarbital anesthesia, and every effort was made to minimize suffering.

2.2 Cell preparation and serum injection

The preparation of single-cell suspensions from skin tumor tissues involved the use of a cell suspension preparation kit (Lot: KFS439, Beijing Baiaolaibo Technology Co, China), with a slight modification. Briefly, the dorsal skin tumor tissues were washed with PBS and cut into small fragments of 1-2 mm in size in a Petri dish containing EDTA/Trypsin. The minced tumor pieces were then transferred to a tube containing trypsin and incubated at 37°C for an hour with shaking. DMEM/10% FBS was added to the dish to recover all cells and tissue, which were then passed through a 100-mm cell strainer. The cell suspension was centrifuged at 500×g for 5 min, and the recovered cells were plated out, ideally at densities of 1×10^5 per 100 mm dish in KC growth medium (**Figure 2A**). The cells were then incubated at 37°C in a 5% CO₂ incubator for 7 days with daily medium changes [11]. Each mouse in the tumor + serum treatment group was randomly paired with a mouse in the serum provider group. Approximately 5×10^5 primary tumor cells suspended in PBS were injected into the tail vein of the paired mice in the serum provider group. After 7 days, 0.1 ml of whole blood was collected from the tail vein of the mice in the serum provider group under ether anesthesia. The serum was immediately separated by brief centrifugation, yielding about 0.02-0.05 ml of serum each time. This serum (0.02 ml) was then injected into the tail vein of its paired mouse in the tumor + serum treatment group once a week, for a total of three times (from week 15 to 17).

2.3 Enzyme linked immunosorbent assay

Previous research has established a connection between the levels of p53, Bcl-xL, NF-κB, and Bax and the occurrence, progression, and metastasis of mCSCC [12-15]. Consequently, this study measured the concentrations of p53, Bcl-xL, NF-κB, and Bax in tissue samples using an ELISA assay. The ELISA Development Kits for mouse p53 (Lot: ab224878), Bcl-xL (Lot: ab227899), NF-κB (Lot: ab176648), and Bax (Lot: ab233624) were procured from Abcam, China. The procedure was as follows: The coated antibody was diluted and added to the ELISA plate (100 μL/well) and incubated for 48 hours at 4 °C. The ELISA plate was then washed three times with tris-buffered saline (TBS) and the diluted sample (100 μL/well) was added and incubated for 90 minutes at 37 °C. After washing three times, all samples were incubated with the diluted enzyme-labeled antibody (100 μL/well) for 60 minutes at 37 °C. The plate was washed three times again, and then the avidin-biotin-peroxidase complex (ABC) developer (100 μL/well) was added. After a 30-minute incubation in the dark at 37 °C, the reaction was stopped using 100 μL of stop buffer. Finally, the plates were read at 450 nm on a microplate reader (Thermo, China).

2.4 Statistical analysis

Data are presented as mean± standard deviation (SD) from three independent experiments. Differences before and after treatment were analyzed using paired sample *t*-test with SPSS 16.0 software package (SPSS Inc., Chicago). *P* value less than 0.05 was considered statistically

Figure 1.

The workflow of this study.

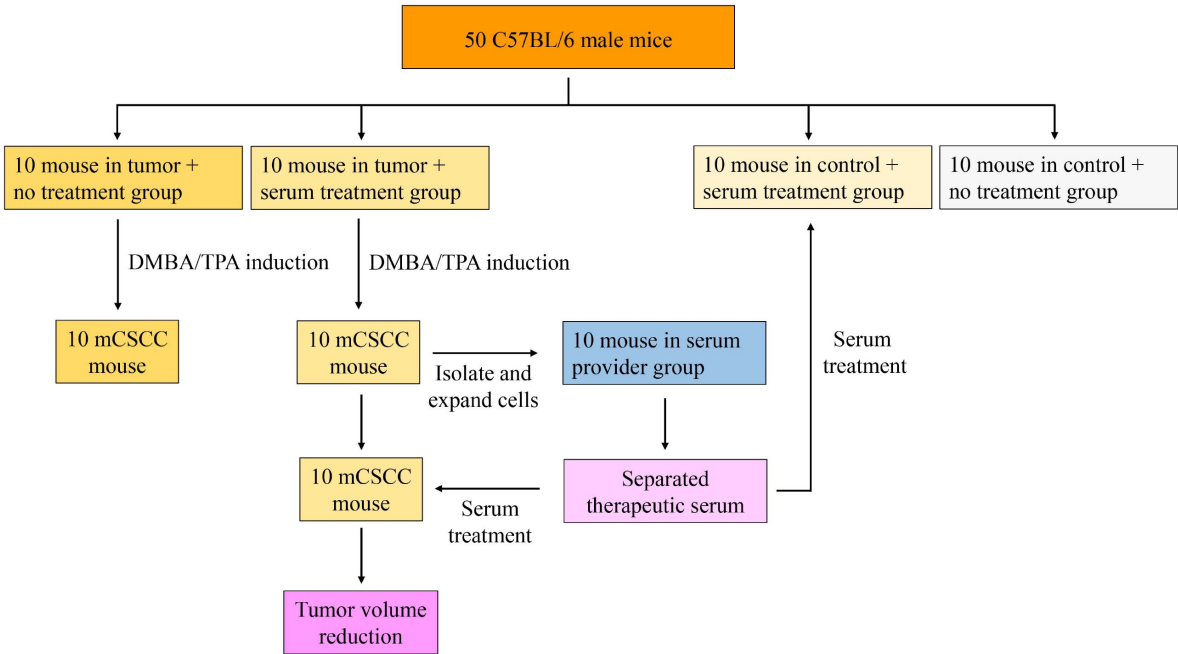
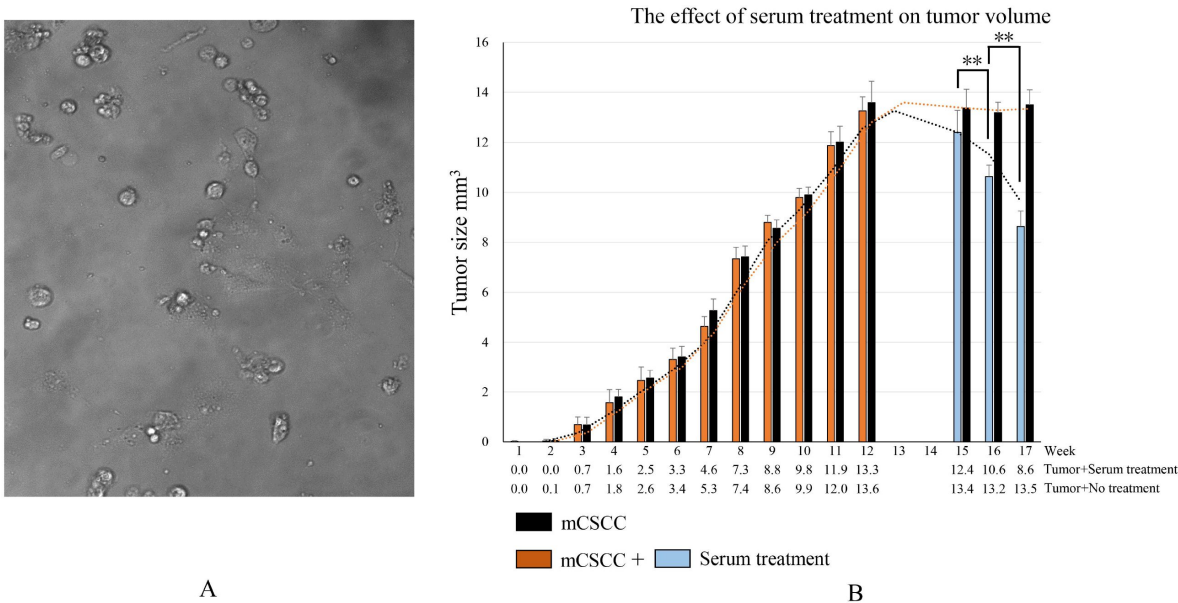


Figure 2.

A. Tumor cells isolated and cultured from mouse cutaneous squamous cell carcinoma (mCSCC). B. Tumor growth induced by DMBA/TPA and changes in tumor volume before and after serum treatment. At week 12, the tumor volume reached its peak. Week 13 was dedicated to the isolation and expansion of the tumor cell. In week 14, the tumor cells were injected into the tail vein of paired mice in the serum provider group to produce serum-based antibodies. Weeks 15, 16, and 17 represent mice in tumor + serum treatment group receiving three times of serum treatment, respectively.

$**P < 0.01$



significant. All experiments were repeated at least three times.

3. Results

3.1 Monitoring body weight

At the beginning of the experiment, male C57BL/6 mice (aged 6-8 weeks) had an average body weight of 20.5 ± 0.3 g, with a range of 20.0 to 21.3 g. Following 12 weeks of DMBA/TPA treatment, mCSCC developed on the backs of these mice. The average body weights of the DMBA/TPA-treated and control animals were 24.9 ± 1.1 g and 26.7 ± 0.8 g, respectively. At the end of experiment (week 17), the average body weights were as follows: 26.6 ± 1.4 g for the tumor + serum treatment group, 27.6 ± 1.2 g for the tumor + no serum treatment group, 28.5 ± 0.8 g for control 1 group (control + serum treatment), and 28.5 ± 0.8 g for control 2 group (control + no serum treatment).

3.2 Serum treatment inhibits the growth of mCSCC

During the DMBA/TPA induction phase, the tumor progressively grows, reaching its peak average volume at 12 weeks. This volume measures 13.3 mm^3 in tumor + serum treatment group and 13.6 mm^3 in tumor + no serum treatment group. In the group that did not receive serum treatment, no significant changes in tumor volume (13.5 mm^3) were observed by week 17. However, after 3 weeks of serum treatment, the tumor volume dramatically reduced to 8.6 mm^3 in tumor + serum treatment group. This substantial decrease demonstrates the efficacy of serum treatment in reducing tumor volume (**Figure 2B** [↗](#)).

3.3 Serum treatment reverses the expression of cancer biomarkers

The ELISA assay results indicate that in mCSCC, the expression levels of p53, Bcl-xL, and NF- κ B are high, while Bax is expressed at a lower level. However, following serum treatment, the levels of p53, Bcl-xL, and NF- κ B decreased, whereas the expression of Bax increased (**Figure 3** [↗](#)). These findings suggest that serum treatment can effectively reverse the expression of cancer biomarkers.

4. Discussion

The principle behind developing this immunotherapeutic strategy is to treat various stages and types of tumor cells in the tumor mass as a whole entity. The various mutated proteins on tumor cells would be sensitively recognized as foreign objects and generating corresponding antibodies in a healthy individual [[16](#) [↗](#)]. Although some tumor cells may evade the patient's immune system, they still can stimulate the production of serum-based antibodies in healthy mice. This strategy is divided into three stages: isolating tumor cells, producing serum-based antibodies, and eliminating the tumor cells (reducing tumor volume). After isolating the tumor cells, the cells of various growth stages and types in a culture medium were expanded.

Injecting these cells into healthy mice led to the production of thousands of antibodies against the corresponding antigens (mutated proteins) on the tumor cells. The serum from the blood of these healthy mice was then transfused back into the tumor-bearing mice to treat mCSCC (**Figure 4** [↗](#)). Given that different stages of tumor cells have distinct surface biomarkers [[17](#) [↗](#)], the serum treatment procedure were repeated weekly for a total of three times (from week 15 to 17). The findings revealed a significant reduction in the tumor volume of the mice. To validate this treatment strategy, p53, Bcl-xL, NF- κ B, and Bax, four mCSCC-associated proteins were selected as tumor biomarkers. In mCSCC, there was a notable increase in p53, Bcl-xL, and NF- κ B, and a decrease in Bax. Serum antibodies for p53, Bcl-xL, NF- κ B, and Bax were produced after injecting tumor cells into healthy mice. The tumor volume decreased following the serum treatment, which

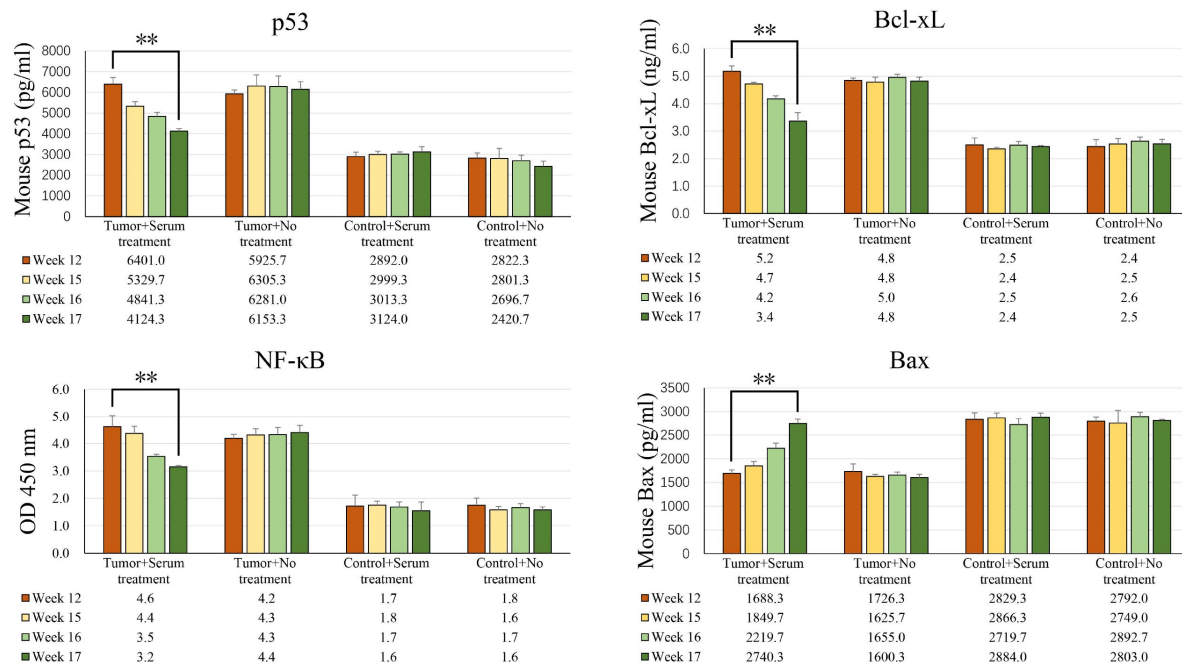


Figure 3.

ELISA analysis revealed changes in the expression of p53, Bcl-xL, NF-κB, and Bax proteins before and after serum treatment. The tumor volume reached its peak at week 12. The mice in tumor + serum treatment group received serum treatment at weeks 15, 16, and 17, respectively.

** $P < 0.01$

was accompanied by a reversed change in p53, Bcl-xL, NF- κ B, and Bax levels. Regrettably, one healthy mouse from the serum provider group and one tumor mouse that received serum treatment died during the study due to unknown reasons.

As early as 1973, research demonstrated that the transfer of serum antibodies could decelerate tumor growth [18]. However, following this discovery, research on serum therapy for tumors nearly halted. In recent years, with a deeper understanding of antibodies and immune cells, immunotherapy has emerged as a significant area of interest [19]. Despite this, the primary focus of research has been on T cells, with B cells receiving less attention. B cells play a crucial role in tumor development and treatment. Upon encountering antigens such as mutated protein, B cells secrete antibodies [20]. Tumorigenesis is a complex and dynamic process. As tumor cells start to develop, the structure of certain proteins changes due to mutations within these cells. These altered proteins can be recognized as non-self-antigens. However, some of these cells gradually adapt and manage to evade the body's immune system [21]. When the number of tumor cells surpasses a certain threshold, a tumor starts to form.

Throughout this process, proteins within the tumor cells continuously accumulate various mutations to adapt to the immune system and the microenvironment [22]. Different epitopes on the mutated protein are exposed on the surface of tumor cells at various stages of the tumor. Occasionally, there is random exposure of these epitopes. This variability in epitope exposure is the primary reason for the immune system's inability to target the tumor effectively, leading to tumor immune escape and the failure of targeted drug treatments. In this study, we propose that these epitopes on the mutated proteins can be recognized as foreign objects, triggering an immune response and generating antibodies in a healthy individual. This leads to an intriguing question: Can B cells be leveraged in a healthy individual to generate anti-tumor antibodies from tumor-exposed individual for cancer treatment? The proposed treatment mechanism involves these diverse antibodies binding to the corresponding epitopes of mutated proteins on the cancer cells. This binding could block cancer cell growth or activate signaling pathways, leading to cell death or apoptosis through antibody-dependent cell cytotoxicity. Moreover, we posit that solely living organisms have the capacity to generate a vast array and diversity of both known and unknown anti-tumor antibodies targeting these epitopes on the mutated proteins. The findings of this study substantiate the hypothesis. Currently, humans have the capability to synthesize certain known neoantigens to help immune system launch the strongest attack against the tumor, such as the Moderna's cancer vaccine mRNA-415, which consists of a single synthetic mRNA coding for up to 34 neoantigens [23]. However, as previously discussed, these artificially created antigens may lead to a situation similar to targeted therapy, where some tumor cells evade immune elimination.

This study has several limitations as follows: (1) The anti-tumor antibodies need to be identified. However, current methods for identifying both known and unknown antibodies pose a significant challenge. (2) Investigating immune response factors in the serum, such as cytokines, is crucial. However, there is a concern that the overall therapeutic effect may be compromised if antibodies and cytokines are considered separately. This concern stems from numerous studies, including those in traditional drug research, which have encountered failures when such a separation was attempted. The fundamental reason for this is that antibodies and cytokines form a mutually activating network. (3) Whole blood therapy may prove to be more effective due to the presence of immune cells such as T cells, B cells, and NK cells. These antibodies, cytokines, and immune cells form an interactive network that collaboratively works towards tumor reduction. However, while studying these components individually, it is crucial to consider the overall therapeutic effect. Furthermore, antibodies and cytokines that are either unknown or present in low concentrations should not be overlooked. (4) The impact of tumor cells on healthy mice is a critical factor to consider. The introduction of exogenous cells into the bodies of healthy mice may result in unpredictable outcomes [24]. Two mice succumbed with weight loss, necessitating further investigation into the causes of this occurrence. Furthermore, is it possible for exogenous tumor cells to trigger immune storms or induce tumor formation in recipient mice? (5) In this study, only

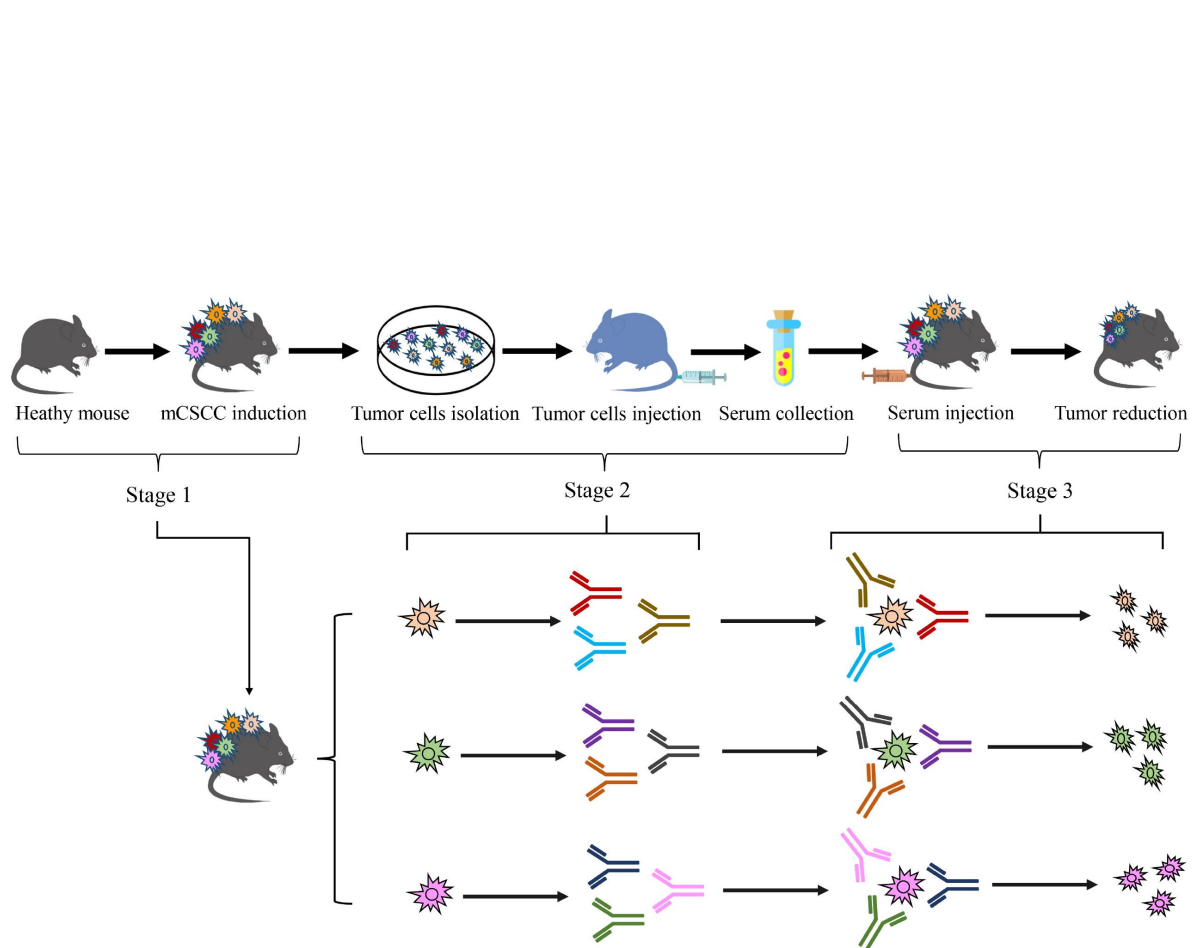


Figure 4.

Schematic diagram of experimental design.

the blood type differences (type A or type B) of mice were considered, without taking into account other factors such as histocompatibility [25]. This is why paired mice were used in this study to reduce side effects. However, employing a completely random process for allocating the treatment groups would be preferable. (6) Although this treatment method has proven successful in mice, additional experiments are necessary before it can be applied to humans. For instance, the current ethical guidelines prohibit the injection of exogenous cells into the human body for the production of therapeutic serum. The complexity of the human body far exceeds that of mice, making it crucial to determine the appropriate dosage of tumor cells, the quantity of anti-tumor antibodies produced, and whether shortening or extending the duration of cell expansion (currently 7 days) or serum-based antibody production (also 7 days) would be more effective. (7) The question arises whether it would be beneficial to use serum treatments with antibodies derived from different animals. While this approach could potentially enhance treatment outcomes, it also introduces new challenges such as the selection of suitable animals, issues related to xeno-transplantation, and managing cross-species immune responses.

In conclusion, this research has explored a new strategy for mCSCC treatment by generating serum-based antibodies from tumor-exposed mice. The method involved the isolation of mCSCC cells, which were subsequently injected into healthy mice. This process stimulated the production of various anti-tumor antibodies present in the serum. These serums were then reintroduced into the tumor-bearing mice, effectively reducing the tumor volume. This cancer treatment method is very effective in treating mCSCC. However, certain aspects of the experiment warrant further investigation and resolution.

Funding

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Conflicts of Interest

The authors declare no conflicts of interest.

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Combined Public Reviews:

Summary:

This study presents an immunotherapeutic strategy for treating mouse cutaneous squamous cell carcinoma (mCSCC) using a passive immunity-like strategy. The researcher induced tumors in healthy mice skin, then isolated the tumor cells and injected into other healthy mice to produce anti-tumor antibodies, and then administered these antibodies back into tumor-bearing mice. Results showed a reduction in tumor volume and altered expression of several cancer markers (p53, Bcl-xL, NF- κ B, Bax). The analysis of results suggests a promising impact of antibody-rich serum in treating mouse cutaneous squamous cell carcinoma (mCSCC).

Strengths:

The approach does seem to have effect on preventing tumor progression, from both the tumor size and the cancer hallmarks expression level.

Weaknesses:

Despite the strength of the study, there are a few drawbacks in the study design and statistical analysis:

(1) Regarding the statistical analysis, the use of a paired t-test might be suboptimal for assessing the trend from weeks 15 to 17. It is recommended to consider alternative methods such as repeated measures ANOVA or linear regression to better capture and interpret the trend over this time period.

(2) To affirm the antibodies' role in the observed immune response, isolating antibodies rather than employing whole serum could provide more conclusive evidence. Comparative analyses with antibody-free serum or serum from healthy, non-immunized mice would clarify antibodies' specific contributions versus other serum components. The control group does not account for the potential immunostimulatory effects of serum injection itself. A better control would be tumor-bearing mice receiving serum from healthy non-mCSCC-exposed mice.

Response to author's rebuttal:

I acknowledge the value of evaluating serum therapy as a whole, considering the complex interactive networks and potential synergies involved. However, to scientifically understand and assess serum therapy, it remains essential to decompose the serum and identify the effective components. This decomposition would allow for a comparison of individual components with the overall effectiveness, thereby elucidating any synergistic effects. While I agree that identifying specific epitopes and paratopes is indeed challenging and may exceed the scope of academic research, the use of methods such as Protein A purification or other techniques to isolate antibodies and cytokines from the serum is both necessary and feasible. This approach would enable a more detailed analysis of the individual effects of these components. I understand that the authors might not have that much resource, and I acknowledge this limitation. Nonetheless, other than this aspect, I believe the authors have adequately addressed my other concerns.

<https://doi.org/10.7554/eLife.95678.2.sa1>

Author response:

The following is the authors' response to the original reviews.

eLife assessment

This study provides a useful strategy for treating mouse cutaneous squamous cell carcinoma (mCSCC) with serum derived from mCSCC-exposed mice. The exploration of serum-derived antibodies as a potential therapy for curing cancer is particularly promising but the study provides inadequate evidence for specific effects of mCSCC-binding serum antibodies. This study will be of interest to scientists seeking a novel immunotherapeutic strategy in cancer therapy.

Joint Public Review:**Summary:**

This study presents an immunotherapeutic strategy for treating mouse cutaneous squamous cell carcinoma (mCSCC) using serum from mice inoculated with mCSCC. The author hypothesizes that antibodies in the generated serum could aid the immune system in tumor volume reduction. The study results showed a reduction in tumor volume and altered expression of several cancer markers (p53, Bcl-xL, NF-κB, Bax) suggesting the potential effectiveness of this approach.

Strengths:

The approach shows potential effect on preventing tumor progression, from both the tumor size and the cancer biomarker expression levels bringing attention to the potential role of antibodies and B cell responses in cancer therapy.

We greatly appreciate your positive feedback on our study.

Weaknesses:

These are some of the specific things that the author could consider to strengthen the evidence supporting the claims in their study.

(1) The study fails to provide evidence of the specific effect of mCSCC-antibodies on mCSCC. The study utilized serum which also contains many immune response factors like cytokines that could contribute to tumor reduction. There is no information on serum centrifugation conditions, which makes it unclear whether immune components like antigen-specific T cells, activated NK cells, or other immune cells were removed from the serum. The study does not provide evidence of neutralizing antibodies through isolation, analysis of B cell responses, or efficacy testing against specific cancer epitopes. To affirm the specific antibodies' role in the observed immune response, isolating antibodies rather than employing whole serum could provide more conclusive evidence. Purifying the serum to isolate mCSCC-binding antibodies, such as through protein A purification, and ELISA would have been more useful to quantify the immune response. It would be interesting to investigate the types of epitopes targeted following direct tumor cell injection. A more thorough characterization of the antibodies, including B cell isolation and/or hybridoma techniques, would strengthen the claim.

I am deeply appreciative of the reviewer's highly professional comments. Tumor development involves the coexistence of cancer cells at different developmental stages, each harboring a variety of known and unknown mutated proteins. These mutated proteins expose multiple known and unknown epitopes, each capable of stimulating the production of corresponding antibodies in healthy mice. Identifying all these antibodies presents a significant challenge. Current research methodologies, such as ELISA, WB, and ChIP, can only identify known antibodies based on existing antigens. A prerequisite for using these techniques is that both antigens and antibodies are identified. At present, there is no

technology available to identify antibodies produced by an unknown mutated protein and epitope. However, I find the reviewer's comments insightful. Perhaps we can initially identify some known mCSCC-antibodies on mCSCC. However, studying the specific effect of these known mCSCC-antibodies on mCSCC is uncertain because we believe that tumor shrinkage results from the combined action of both known and unknown antibodies.

We concur with the reviewer's observations regarding the use of serum, which is rich in immune response factors such as cytokines that could potentially contribute to tumor reduction. In our future research, we plan to systematically analyze the individual roles of these antibodies and cytokines in tumor reduction. In 1973, *Nature* published a report indicating that serum demonstrated promising results in tumor treatment (Immunotherapy of Cancer with Antibody in Rats. *Nature* 243, 492 (1973). <https://doi.org/10.1038/243492b0>). Since then, there have been scarcely any reports on serum therapy for tumors. The primary focus of our study is to evaluate the efficacy of serum therapy in treating tumors. We hypothesize that antibodies and cytokines form a complex interactive network, working in synergy to reduce tumors. Consequently, we believe that studying these antibodies and cytokines in isolation may not yield effective results.

In this study, the methodology section outlines the process of serum preparation. It is important to note that serum is devoid of blood cells. I hypothesized that whole blood might have superior therapeutic effects compared to serum. This is because antibodies could potentially synergize with immune cells (including T cells, B cells, and NK cells), thereby enhancing the effectiveness of the treatment. As previously discussed, these antibodies, cytokines, and immune cells form a complex interactive network aimed at tumor reduction. Consequently, there are numerous factors that could influence the experimental outcomes, which presents a challenge for analyzing the results. Furthermore, the implementation of whole blood transfusion therapy introduces additional considerations, such as potential side effects and reactions associated with blood transfusions.

We thank the reviewers for their suggestion to purify the serum in order to isolate mCSCC-binding antibodies. As we previously mentioned, separating a large number of both known and unknown serum antibodies presents a significant technical challenge. We are eager to discuss and consider suggestions from the reviewers regarding methods to identify a large variety and number of unknown antibodies on cells. Perhaps, as the reviewer suggested, we could begin with known antibodies and employ Protein A purification technology to purify these antibodies and subsequently detect immune responses. We could also categorize the types of epitopes targeted, direct tumor cell injection, to study the epitopes of these types in further studies. The suggestion to study the response of B cells is valuable, and we plan to conduct comprehensive research on the response and status of B cells in our future studies.

The purification of antibodies to enhance the specificity of their effectiveness against tumors is a critical aspect of our study. However, we would like to address some concerns raised. (1) The separation of all antibodies and cytokines presents a significant technical challenge. Particularly, there is a risk of overlooking antibodies that are present in low concentrations but play crucial roles. (2) What concerns us is that studying the composition separately would lose the overall effectiveness of the study. Our primary concern is that studying these components in isolation could compromise the holistic understanding of the study. This is akin to current research on traditional medicine, where the separation and individual study of compounds often result in a loss of overall therapeutic efficacy. For instance, consider a scenario where 100 antibodies collectively work to shrink a tumor. These antibodies interact with 20 cytokines, forming a complex network that enhances the cytokines' activity against tumor cells. Furthermore, many important antibodies and cytokines are currently unknown. Studying these antibodies in isolation could potentially result in the loss of this therapeutic effect. Therefore, in the discussion section, we have emphasized that our study considers a tumor mass, including tumor cells at various stages of development, as a single entity. As a

practicing clinician, my primary focus is on the therapeutic outcomes in tumor treatments, despite the mechanisms of serum therapy remaining largely elusive, liking a black box.

(2) In the study design, the control group does not account for the potential immunostimulatory effects of serum injection itself. A better control would be tumor-bearing mice receiving serum from healthy non-mCSCC-exposed mice. Additionally, employing a completely random process for allocating the treatment groups would be preferable. Also, the study does not explain why intravenous injection of tumor cells would produce superior antibodies compared to those naturally generated in mCSCC-bearing mice.

I concur with the reviewer's perspective that using serum from healthy, non-mCSCC exposed mice as a control could potentially improve our study. Initially, our primary concern was to minimize harm to the mice and avoid excessive blood reactions, which led us to exclude the use of serum from healthy, non-mCSCC exposed mice in our control group. The main objective of our study was to investigate tumor shrinkage through serum treatment, specifically serum-derived antibodies. We anticipated that tumor-bearing mice receiving serum from healthy, non-mCSCC exposed mice would exhibit a response to the injected serum, which would manifest as a blood reaction. However, we did not expect this to result in a tumor treatment effect. If it turns out that normal serum (from healthy, non-mCSCC-exposed mice) possesses tumor-reducing properties, it would indeed be a novel discovery. We appreciate the reviewer's insightful suggestion and will consider incorporating it into our future research.

We concur with the reviewer's observations that the use of a completely random process for assigning treatment groups would be more desirable. Indeed, the complete randomization of the entire process further underscores the efficacy and universality of serum therapy. In this study, we utilized paired mice to mitigate the risk of cross-infection and adverse reactions associated with blood transfusions. We deeply value the reviewer's expert feedback.

Lastly, the reason why tumor cells, when intravenously injected, produce antibodies superior to those naturally generated in mCSCC-bearing mice, is due to the following reasons. As tumor cells grow, they produce a variety of mutated proteins to adapt to the immune microenvironment and evade the immune system of mCSCC-bearing mice. However, these tumor cells with mutated proteins are exceptionally sensitive and recognizable to healthy mice. This recognition triggers an immune response in healthy mice, leading to the production of specific therapeutic antibodies. This simultaneous production of diverse and abundant antibodies is only achievable by living organisms.

(3) In Figure 2B, it would be more helpful if the author could provide raw data/figures of the tumor than just the bar graph. Similarly in Figure 3, the author should show individual data points in addition to the error bar to visualize the actual distribution.

Raw data (numerical values) have been incorporated into Figures 2B and 3, but the data is placed in the table below the graph. If placed above the error bar, it requires a small font and may not be clear.

(4) The author mentioned that different stages of tumor cells have different surface biomarkers. Therefore, experimenting with injecting tumor cells at various stages could reveal the most immunogenic stage. Such an approach would allow for a comparative analysis of immune responses elicited by tumor cells at different stages of development.

Yes, throughout the course of tumor development, tumor cells at various stages will exhibit distinct markers or possess different mutated proteins. The concept of segregating tumor cells from different stages and independently comparing their immune responses is indeed

commendable. Future research could involve isolating cells that express identical biomarkers at each stage for a comparative analysis of the immune responses triggered by the tumor cells. However, this approach diverges from the original intent of this study.

Most tumor cells exist within the same developmental stage. However, this does not imply that all tumor cells within the tumor mass are at the same stage. For instance, a stage III liver cancer tumor may contain both stage I and stage IV tumor cells. Moreover, due to the complexity of tumor development, not all tumor cell surface markers are identical, even for tumors at the same stage. For instance, 20 major proteins and 100 minor proteins are implicated in tumor formation. In fact, random mutations in just 5 of these major proteins and 10 minor proteins can instigate the development of tumors. This implies that the protein pattern (tumor cell surface markers) associated with each individual's tumor is unique. While studying tumor cells at different stages separately allows for the observation of the immune response of tumor cells at each stage, it lacks a comprehensive research and treatment effect. For this reason, the design of this study treats a tumor mass as a whole, encompassing both the primary stage tumor cells and those not in that stage. These tumor cells are then injected to produce corresponding therapeutic antibodies. Furthermore, if tumor cells from only one stage are isolated and specific antibodies are produced against these cells, it could lead to immune escape of tumor cells at other stages, preventing the tumor from shrinking. Therefore, our approach aims to address this issue by considering the tumor mass as a whole.

(5) In the abstract the author mentioned that using mCSCC is a proof-of-concept for this potential cancer treatment strategy. The discussion section should extend to how this strategy might apply to other cancer types beyond carcinoma.

We have incorporated an additional paragraph in the discussion section where we delve into the concepts and experimental principles underpinning this study. This, we believe, addresses the reviewer's query regarding the applicability of our study's methodology to other types of tumors. The process for other tumors also involves isolating cells from the tumor, stimulating therapeutic antibody production in healthy mice using these cells, and ultimately reintroducing these antibodies into mice with tumors to facilitate tumor elimination

Recommendations For The Authors:

The author is encouraged to refine the study's design in future studies considering the weaknesses highlighted above, summarize the results more effectively, and seek opportunities to expand on this promising idea and enhance the research's impact and applicability.

We greatly appreciate the valuable suggestions provided by the editor and reviewers. These insights will certainly be addressed in our future research endeavors.

Suggestions for title modification:

Following the scope of the study, the term 'specific homologous neutralizing-antibodies' may be misleading as neutralizing antibodies typically refer to antibodies preventing viral cell entry. In cancer therapy, 'neutralization' is not a relevant concept, as cancer cells do not infect host cells. Using whole tumor cells as immunogens diverges from the specificity of traditional vaccination approaches that utilize well-defined proteins or antigens. Furthermore, the term "homologous" suggests a precision in targeting that is not demonstrated by reintroducing serum without isolating its specific components. Therapeutic effects should not be attributed to "neutralizing antibodies" without isolating or characterizing the antibody response or verifying their efficacy against specific cancer epitopes. Additionally, it is suggested that you indicate the biological system that your

study utilised in the title. More so, this approach is not entirely novel, as seen with the use of adjuvants in some flu vaccines, or in Moderna's cancer vaccine mRNA-4157, which encodes up to 34 patient-specific tumor neoantigens. You can consider the title below or a variant of the same.

Suggested title: Generating serum-based antibodies from tumor-exposed mice: a potential strategy in cutaneous squamous cell carcinoma treatment

I concur with your suggestion and have modified the title to "Generating serum-based antibodies from tumor-exposed mice: a new potential strategy for cutaneous squamous cell carcinoma treatment". I believe this research remains some new, hence the addition of the word "new". Furthermore, the term "novel" in the paper has been either removed or substituted.

Moreover, I propose that this study shares similarities with Moderna's cancer vaccine mRNA-415, albeit with certain differences. Moderna's cancer vaccine mRNA-415 encodes 34 recognized neoantigens to stimulate an immune response by eliciting specific T cell responses. This is similar to the strategy of some companies developing a protein set for diagnosing lung cancer, liver cancer, among others. Without a doubt, these methods have improved the effectiveness of tumor diagnosis and treatment. However, I think that these methods currently face challenges in completely eradicating tumors because they perceive tumors as a static process and cells that express certain mutated proteins in a fixed manner. I believe that small molecule antibodies, cytokines, and immune cells present in serum that are difficult to detect, have low concentrations, or are unknown are essential for maintaining the expression of important mutant proteins and the escape of tumor cells. This is also the primary reason why tumors are difficult to treat and prone to recurrence at present.

From my perspective, different tumors, as well as different stages of the same tumor, express varying mutated proteins or surface markers. Targeting some may result in others escaping or even creating a more conducive growth environment for those that do escape. Our study adopts a comprehensive view of a tumor block, encompassing tumor cells at different stages and tumor cells at the same stage but expressing different biomarkers. This approach generates a multitude of known and unknown antibodies that work in concert with cytokines and immune cells. While our method may not be capable of generating all mutated proteins and epitope antibodies due to the weakness of some antigens (epitopes of mutated proteins), it can still be effective. As long as the number of tumor cells is reduced below a certain threshold following multiple rounds of treatment with various antibodies produced at different stages, these cancer cells can be eradicated by the body's immune system. This is a process that is real-time and dynamic. Undoubtedly, if it becomes evident that alterations in a set of proteins can bolster the immune system and eradicate tumor cells, then the implications are significant. The immunotherapy proteins, which have demonstrated positive therapeutic effects, developed by certain companies are also predicated on this very principle.

Finally, I greatly appreciate your suggestions, which will be considered and gradually addressed in future research.

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